ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PRACTICE EXAM 2

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 8 year: ratio, rate, VAT and discount, integers and exponents, fractions, patterns, expanding and simplifying algebraic expressions, solving equations, angle relationships, isosceles and similar triangles, Pythagoras, circles, volume and surface area, rotations and enlargements, and data from a frequency table.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone will not get full marks.
4. A non-programmable calculator may be used, unless a question says otherwise.
5. Round answers to TWO decimal places where necessary. Diagrams are NOT drawn to scale.
6. Give reasons for your statements in geometry questions.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 $-15 \div (-3) = \ldots$

A -5 B 5 C 45 D -45

1.2 $2^5 \times 2^3 = \ldots$

A 2^{15} B 4^8 C 2^8 D 4^{15}

1.3 0,6 written as a common fraction in its simplest form is …

A $\frac{6}{100}$ B $\frac{3}{5}$ C $\frac{6}{10}$ D $\frac{2}{3}$

1.4 The next term in the pattern 2 ; 6 ; 18 ; 54 ; … is …

A 72 B 108 C 162 D 216

1.5 If $x = -4$, then $x^2 = \ldots$

A -16 B 16 C -8 D 8

1.6 The interior angles of a quadrilateral add up to …

A 180° B 270° C 360° D 540°

1.7 Two angles that add up to 90° are called …

A supplementary angles. B complementary angles.
C vertically opposite angles. D corresponding angles.

1.8 The volume of a cube with sides of 4 cm is …

A 12 cm³ B 16 cm³ C 64 cm³ D 96 cm³

1.9 $\sqrt{(3^2 + 4^2)} = \ldots$

A 7 B 5 C 25 D 12

1.10 A letter is chosen at random from the word MATHS. The probability that it is a vowel is …

A $\frac{1}{5}$ B $\frac{2}{5}$ C $\frac{1}{4}$ D $\frac{4}{5}$

[10]

QUESTION 2: RATIO, RATE AND FINANCE

2.1 R2 400 is shared among Ayanda, Bongi and Chris in the ratio 2 : 3 : 5. How much does each person receive? (3)

2.2 A car uses 7,5 litres of petrol per 100 km. Petrol costs R23,40 per litre. (a) How many litres does the car use on a trip of 360 km? (b) Calculate the cost of the petrol for the trip. (3)

2.3 A tablet costs R600 before VAT. (a) Calculate the price including 15% VAT. (b) During a sale, the price including VAT is reduced by 20%. Calculate the sale price. (4)

[10]

QUESTION 3: INTEGERS AND EXPONENTS

3.1 Calculate WITHOUT a calculator: (a) $(-6)^2 \div (-4) + (-3)^3$ (b) $\sqrt{144} - \sqrt[3]{(-125)}$ (5)

3.2 Simplify: (a) $a^5 \times a^3 \div a^4$ (b) $(-3x^2y)^2$ (c) $(2x)^3$ (5)

[10]

QUESTION 4: FRACTIONS AND DECIMALS

4.1 Calculate WITHOUT a calculator. Give answers in the simplest form: (a) $3\frac{1}{4} - 1\frac{5}{6}$ (b) $2\frac{2}{3} \div 1\frac{1}{3}$ (c) $\sqrt{(\frac{25}{36})} + \frac{1}{2}$ (7)

4.2 Round off 3,14159 to three decimal places. (1)

[8]

QUESTION 5: PATTERNS AND ALGEBRA

5.1 Matchsticks are used to build figures. Figure 1 uses 3 matchsticks, Figure 2 uses 5 and Figure 3 uses 7.

5.1.1 Write down the general term for the number of matchsticks in Figure *n*. (2)

5.1.2 How many matchsticks are needed for Figure 50? (1)

5.1.3 Which figure uses 81 matchsticks? (2)

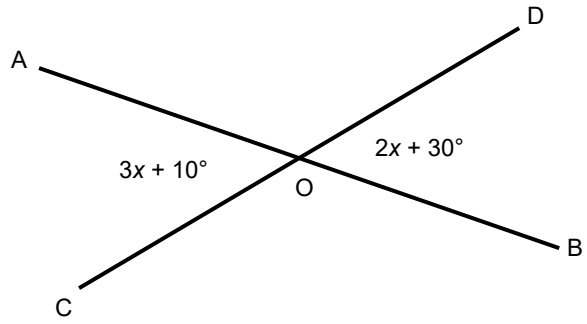
5.2 Simplify: (a) $4(2x - 3) - 3(x - 5)$ (b) $x(x^2 - 2x + 3)$ (c) $(18x^4y^2 - 6x^2y) \div 6x^2y$ (d) $\sqrt{(16x^8)}$ (8)

5.3 Solve for x: (a) $4x - 9 = 2x + 5$ (b) $x - \frac{1}{3} = 4$ (c) $2x^2 = 50$ (6)

[19]

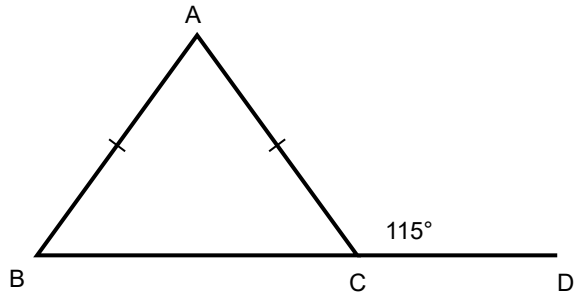
QUESTION 6: GEOMETRY

6.1 AB and CD are straight lines that intersect at O. $\angle AOC = 3x + 10°$ and $\angle BOD = 2x + 30°$.



Calculate x, and then the sizes of ∠AOC and ∠COB. Give reasons. (4)

6.2 In ΔABC, AB = AC. BC is extended to D, and ∠ACD = 115°.



Calculate, with reasons: (a) ∠ACB (b) ∠B (c) ∠A (4)

6.3 AB ∥ CD. A transversal cuts the two lines so that two co-interior angles are x and $2x$. Calculate x. Give a reason. (3)

6.4 ΔABC is similar to ΔDEF (ΔABC ||| ΔDEF). AB = 6 cm, BC = 8 cm and DE = 9 cm. Calculate EF. (2)

6.5 Name the quadrilateral whose diagonals are equal in length and bisect each other at right angles. (1)

[14]

QUESTION 7: PYTHAGORAS AND MEASUREMENT

7.1 A television screen is 80 cm wide and 60 cm high. Calculate the length of its diagonal. (2)

7.2 A right-angled triangle has a hypotenuse of 13 cm and one side of 5 cm. Calculate the length of the third side. (2)

7.3 A circle has a diameter of 20 cm. Use π = 3,14 to calculate: (a) the area of the circle; (b) the perimeter of a semicircle cut from this circle (the curved edge plus the diameter). (4)

7.4 A fish tank (without a lid) is 60 cm long, 30 cm wide and 40 cm high. (a) Calculate its volume in cm³ and in litres (1 000 cm³ = 1 ℓ). (b) Calculate the area of glass needed to make the tank. (6)

[14]

QUESTION 8: TRANSFORMATIONS

8.1 Triangle ABC has vertices A(1 ; 1), B(4 ; 1) and C(1 ; 3).

8.1.1 Give the coordinates of A', B' and C' if ABC is rotated through 180° about the origin. (3)

8.1.2 Give the coordinates of the image of C after a reflection in the *y*-axis. (1)

8.1.3 Give the coordinates of the image of B after an enlargement by a scale factor of 3 with the origin as centre. (1)

[5]

QUESTION 9: DATA HANDLING AND PROBABILITY

9.1 30 learners were asked how many books they read during the holidays.

Number of books	0	1	2	3	4
Number of learners (frequency)	4	7	9	6	4

9.1.1 Write down the mode. (1)

9.1.2 Calculate the total number of books read. (2)

9.1.3 Calculate the mean number of books read per learner. (1)

9.1.4 Determine the median. Show how you got your answer. (2)

9.1.5 Write down the range. (1)

9.1.6 One learner is chosen at random. What is the probability that the learner read more than 2 books? Simplify your answer. (2)

9.1.7 Which type of graph would be best to display this data? (1)

[10]

TOTAL: 100

END OF QUESTION PAPER